

Cashflow Studio Workflow

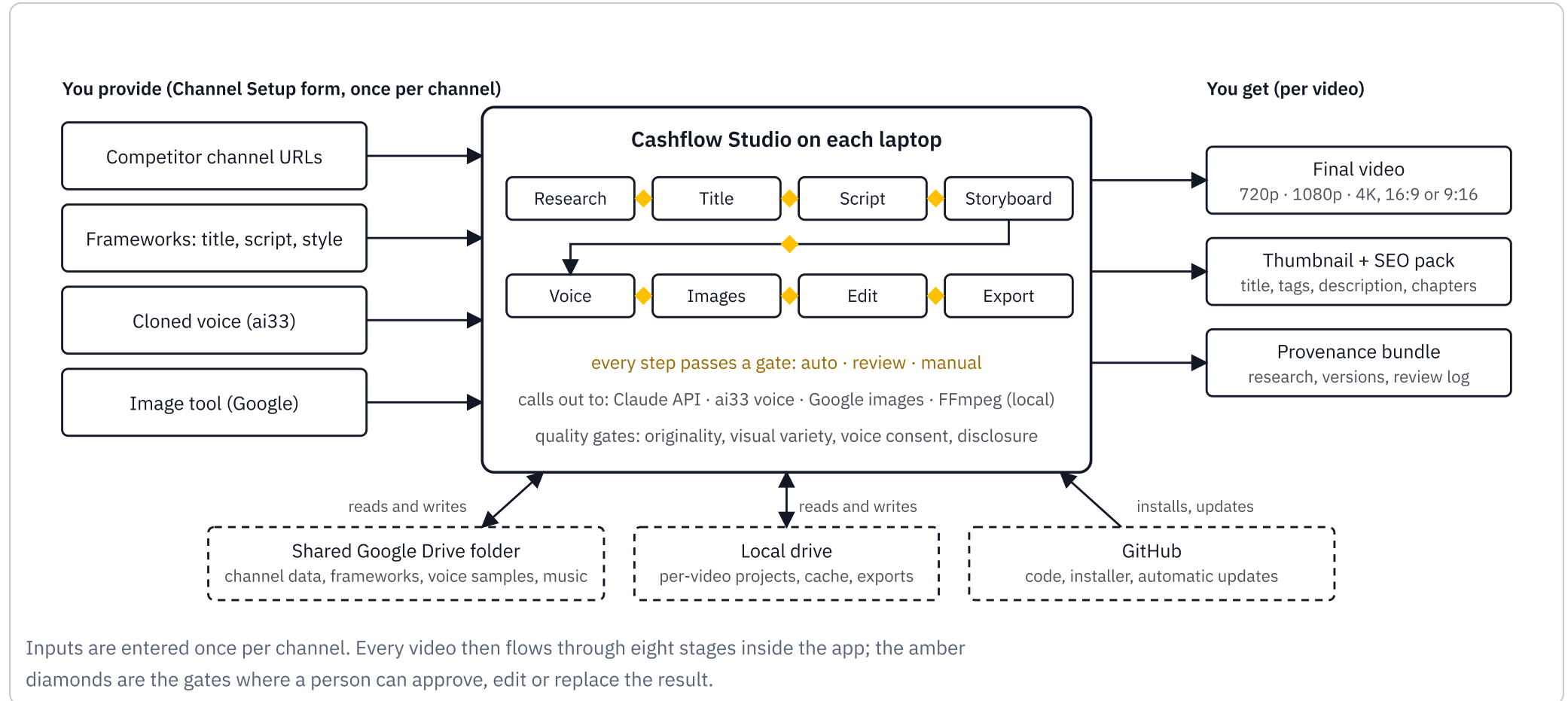
How the system works from a list of competitor channels to a finished, exported video, with a person able to approve, edit or replace the result of every stage. Written for the whole team: editors, reviewers and developers.

VERSION v0.3 · 2026-10-09	VOICE TOOL ai33 (link to be confirmed)	IMAGE TOOL Google (Gemini image models)	WRITING AND CHECKS Claude API (Opus 5.5, Sonnet 5.5)	FORMATS 16:9 long-form and 9:16 Shorts · 720p, 1080p, 4K
RUNS ON Each editor's Windows laptop, no GPU needed				
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SECTION 1

The whole system on one page

Each editor runs the app on their own laptop. Channel data is shared through a synced Google Drive folder; media stays local; code and updates come from GitHub. Every video flows through eight stages, and every step between stages is a gate where a person can act.



◆ gate between stages: auto · review · manual — data flow - - - service or store outside the app

People, components and services

Who and what takes part, and what each one is responsible for.

ACTOR OR COMPONENT	RESPONSIBILITY	WHERE IT RUNS
Editor	Creates videos, reviews stages, fixes storyboards and cuts, approves exports	Own laptop, in the app
Channel owner / reviewer	Sets up channels, frameworks and stage modes; final approvals; policy overrides with a logged reason	Own laptop, in the app
Desktop app	Dashboard, Channel Setup form, Research, Storyboard board, Timeline editor, Review queue, Settings	Native window on the laptop (web technology inside)
Pipeline engine	Runs the eight stages, enforces gates and quality checks, tracks state, costs and provenance per video	Local background service, started by the app
Research engine	Lists competitor videos, scores outliers, fetches transcripts and thumbnails	Local, yt-dlp, read-only access to YouTube
Claude API	Titles, scripts, scene splitting, image prompts, SEO pack, image quality check, transition choice	Cloud, each user's own key
ai33 (voice)	Speech from the cloned voice, per sentence, with word timings where supported	Cloud, the team's account
Google (images)	One text-free image per scene, thumbnails, with a style sheet image for consistency	Cloud, Gemini image models through a Google API key
FFmpeg	Zoom and pan, transitions, captions, popups, music ducking, final encode at 720p, 1080p, 4K	Local, bundled with the app
Shared Drive folder	Channel configs, frameworks, style references, voice samples, music library, thumbnail templates	Google Drive, synced to each laptop
GitHub	Source code, release builds, update manifest	Cloud, developers only

The Google consumer subscription (Flow, the Gemini app) has no programming interface. The pipeline uses the Gemini image models through a Google API key with pay-as-you-go billing, roughly three to five US cents per image. Images made by hand in Flow can still be dropped into any scene through the manual option.

One-time setup per channel

The Channel Setup form collects everything needed to start work. It is saved as one file per channel in the shared folder, so every teammate's app sees the same channel.

Channel

- Name, YouTube URL (the app resolves the channel id)
- Main language, optional secondary languages
- Niche in one line, audience in one line
- Formats: long-form, Shorts, or both; target lengths
- Videos per week (used for batch planning)
- Export folder, music folder, brand colours, caption style
- Owner, browser profile number for upload, status

Competitors

- One row per channel: name, URL
- Priority: main, secondary, watch only
- Why we track it (free text)
- Filled by the app: channel id, videos found, last scanned

Frameworks

- Type: title, long-form script, Shorts script, scene prompt, thumbnail, SEO, style guide
- Uploaded file (PDF, TXT, MD) or link, version, which format it applies to
- The script stage reads the whole framework text into its instructions

Voice

- Tool (ai33), clone link or voice id, voice name
- Language, model or quality setting, speed, style notes
- Sample recording path, whether the tool returns word timings
- Name of the key variable, monthly budget in characters
- Consent record: whose voice, when and how consent was given

Images

- Tool (Google), model, style guide sentence added to every prompt
- Negative rules the quality check rejects
- Aspect ratios for long-form and Shorts, output resolution
- Reference images folder (style sheet), popup text style
- Name of the key variable, monthly budget in images

Thumbnail

- One to three competitor videos whose thumbnail layout is the template
- Headline font, colours, maximum headline words
- Face, no face, or AI character; must include; must avoid
- Output sizes for long-form and Shorts

Stage modes

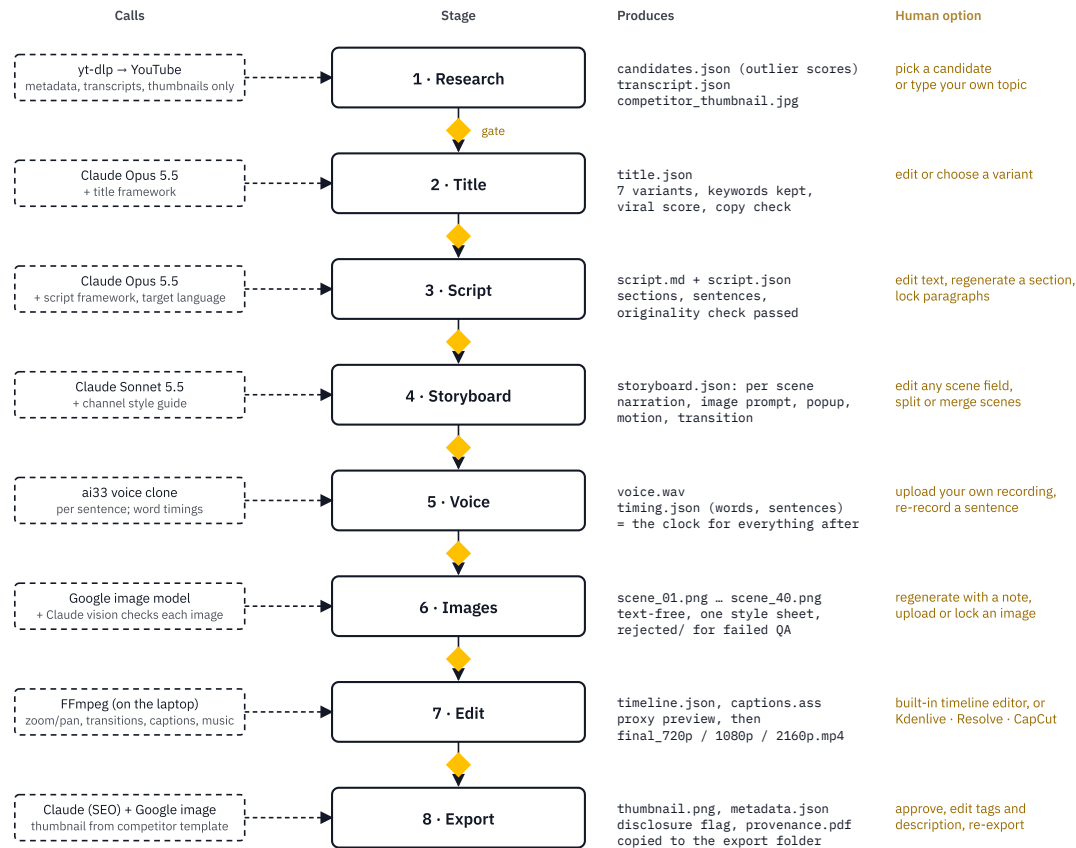
- For each of the eight stages: auto, review or manual
- Reviewer name per stage
- Defaults: everything on review except Voice on auto
- Can be overridden per video when it is created

Settings (per laptop, not per channel)

- Shared folder path, projects folder, exports folder
- API keys for Claude, Google, ai33: stored only on that laptop
- Doctor: checks FFmpeg, yt-dlp, fonts, WebView2, folders, keys, disk space

One video, stage by stage

Read top to bottom. The left column is the service each stage calls; the right column is the human option at that stage. Each gate is set per channel to auto (continue), review (wait for approval) or manual (a person supplies the file).



a failed quality gate (copied script, reused image, missing consent, unfulfilled title) blocks export until a reviewer overrides it with a logged reason

Every stage reads the files of the stage before it and writes its own files into the video's project folder, so a person can open any folder and change anything between stages. Dashed boxes are services outside the app.

Stage specifications

Each card is one stage: what goes in, exactly what the app does, what comes out, what is checked, and what a person may do. Times are estimates for a ten-minute video on a laptop without a GPU.

01 Research		02 Title	
INPUT	Competitor list from the channel; optional topic typed by the editor	INPUT	Chosen candidate title and its metrics; the channel's title framework; the niche and audience lines
STEPS	List each competitor's videos and Shorts tabs without downloading media; cache results; compute an outlier score per video = views divided by the median views of the neighbouring videos of that channel, age-normalised, plus views per subscriber; fetch exact numbers for the top 20 to 50; for the chosen video, fetch the transcript (for structure only, never copied) and the full-size thumbnail	STEPS	Claude writes 7 variants that keep the proven keywords and structure but are not copies; each gets formula, emotional trigger, curiosity trigger, viral score and a reason it could outperform; a similarity check rejects near-copies of the competitor and of the channel's recent titles
OUTPUT	candidates.json · transcript.json · competitor_thumbnail.jpg	OUTPUT	title.json
CHECKS	Candidate newer than 7 days is excluded from baselines; channel not scanned more than once per hour; back-off when YouTube asks for sign-in	CHECKS	Length under 70 characters, no misleading promise the script cannot keep (re-checked after the script exists)
HUMAN	Pick a candidate, type your own topic, exclude a channel, rescan	HUMAN	Edit, choose, reorder, ask for 7 more
TIME	1 to 3 minutes	TIME	Under a minute

03 Script

INPUT	Approved title; script framework (long-form or Shorts); target language; target length; transcript summary of the competitor video
STEPS	Claude writes the script in sections (hook, setup, progression, payoff, call to action as the framework defines) with sentence boundaries marked; a speech-normalisation pass expands numbers, dates and currency in the target language so the voice reads them correctly; word count set from the target length and the voice's speaking rate
OUTPUT	script.md (readable) · script.json (sections and sentences)
CHECKS	Originality: n-gram and meaning similarity against the competitor transcript and the channel's last 30 scripts must stay under the ceiling; no advisory persona on finance, health or legal topics; title promise appears in the first 20 percent
HUMAN	Edit any paragraph, regenerate one section, lock paragraphs so regeneration never touches them
TIME	2 to 5 minutes plus reading

04 Storyboard

INPUT	Approved script sentences; channel style guide; popup style; aspect ratio
STEPS	Claude groups sentences into scenes (about 8 to 12 seconds each for long-form, 3 to 4 for Shorts); for each scene writes a text-free image prompt in the channel style, chooses the popup phrase (at most 6 words paraphrasing the narration), a motion preset (zoom in, zoom out, pan, hold) and the transition into the next scene; motion alternates so no two neighbouring scenes move the same way
OUTPUT	storyboard.json
CHECKS	Scene density within limits; at least a set share of scenes carry a popup, chart card or text card; no two prompts near-identical
HUMAN	Edit any field, split or merge scenes, reorder, lock a scene
TIME	1 to 2 minutes plus review

05 Voice

INPUT	Approved script sentences; the channel's cloned voice (ai33), speed and style settings
STEPS	Synthesise sentence by sentence (small requests, exact sentence boundaries, cheap re-dos); concatenate with sample-accurate offsets; collect word timings from the tool when it returns them; otherwise align in the background on the CPU (several minutes) or fall back to sentence timing; cache every sentence by a content hash so an edit re-bills only the changed sentence
OUTPUT	voice.wav (48 kHz) · timing.json
CHECKS	Words shorter than 40 ms or longer than 2 s are flagged; total duration against target length; consent record present for the voice
HUMAN	Upload your own recording (aligned automatically), re-record a sentence, nudge a timing
TIME	1 to 3 minutes

06 Images

INPUT	Storyboard prompts; style sheet image; aspect ratio; negative rules
STEPS	Generate one image per scene at the largest size the model offers, passing the style sheet and the previous scene as references for a consistent look; Claude vision checks each image against its prompt (subject present, no garbled text, no real person, no logo, no deformed hands); failures are regenerated up to three times with the reason added to the prompt; a perceptual hash is compared against the channel's past images to prevent reuse
OUTPUT	scene_01.png ... scene_NN.png · rejected/ · style_sheet.png
CHECKS	Quality check verdict stored per image; budget guard warns before a batch exceeds the monthly plan
HUMAN	Regenerate with a note, upload an image (for example one made in Flow), lock an image
TIME	5 to 10 minutes for 40 scenes

07 Edit

INPUT	Storyboard, voice timing, images, music folder, caption style, brand colours
STEPS	Build the timeline: scene in and out from sentence timestamps; zoom and pan path per scene; chosen transition at each boundary; popups rendered as image layers with the brand font; word-level captions as a subtitle track; music under the voice with automatic ducking; render a low-resolution proxy for preview; render the final presets
OUTPUT	timeline.json · captions.ass · popups/ · proxy.mp4 · final_720p.mp4 · final_1080p.mp4 · final_2160p.mp4
CHECKS	Audio peaks, caption readability per language, no black frames, duration equals voice length plus intro and outro
HUMAN	Fix in the built-in editor; or export the project to Kdenlive, DaVinci Resolve or CapCut and import the finished file
TIME	5 to 15 minutes per preset on a laptop

08 Export

INPUT	Final renders; approved title; script; competitor thumbnail template; channel brand settings
STEPS	Thumbnail: a vision pass extracts the competitor layout (subject position, text blocks, colours, shapes) as a template; the image model draws a text-free subject; the app composes headline and shapes on top in the brand font; SEO pack: Claude writes title, description, tags, chapters and the pinned comment in the target language; disclosure flag set to yes by default; provenance bundle assembled; everything copied to the channel's export folder
OUTPUT	thumbnail.png · metadata.json · provenance.pdf · export folder
CHECKS	Thumbnail similarity to the competitor's below the ceiling; title and thumbnail promise found in the script; music licence stored
HUMAN	Approve, edit metadata, swap thumbnail variant, re-export
TIME	1 to 2 minutes

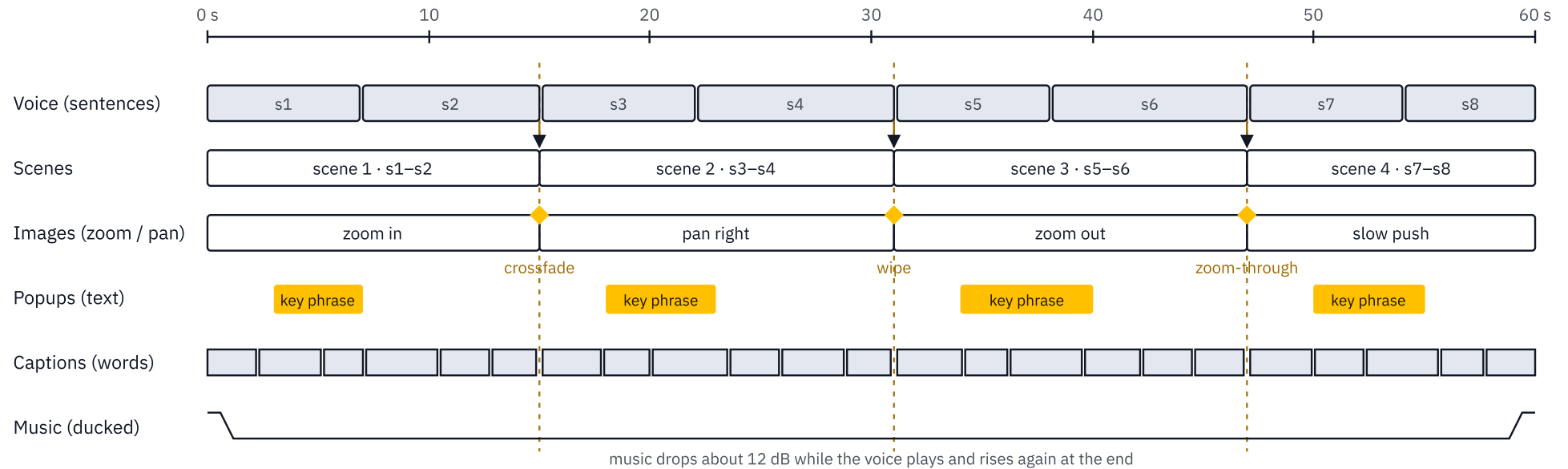
The storyboard record

The storyboard is the centre of the product. Every scene is one record with the fields below; the board screen edits them, the voice stage times them, the image stage fills them, and the edit stage renders them. A locked field is never overwritten by a regeneration.

FIELD	MEANING	SET BY	EDITABLE
index	Order of the scene	Storyboard stage	Reorder, split, merge
narration	The sentences spoken during this scene	Script stage	Yes (re-voices only this scene)
start, end	Seconds on the timeline, from the voice timing	Voice stage	Nudge in the editor
image_prompt	Text-free picture description in the channel style	Storyboard stage	Yes
image	Path of the approved image and its quality verdict	Images stage	Regenerate, upload, lock
popup	Up to 6 words, style preset, in and out times, position	Storyboard stage	Yes
motion	Zoom in, zoom out, pan left or right, hold; start and end frame rectangles	Storyboard stage	Yes, with handles in the editor
transition	Type and duration of the cut into the next scene	Storyboard stage	Yes, picker
captions	Word cues inside the scene	Voice stage	Text fixes
locked	Which fields a person has fixed by hand	Editor	Toggle
notes	Reviewer comments and regeneration reasons	Anyone	Yes

Voice timing drives the edit

The voice carries word and sentence timestamps. Those timestamps set where each scene starts and ends, how long each image stays, when a popup appears, when captions change and when the music ducks. This is what makes the automatic edit look cut to the narration instead of to a fixed clock.



Sentence boundaries from the voice timing (s1 to s8) are grouped into scenes; each scene gets one image with its own motion, a chosen transition at the boundary, a popup for its key phrase, word-level captions, and music that ducks under the voice.

How transitions are chosen

Claude assigns a transition to every cut from FFmpeg's library (crossfade, wipes, slides, circle and radial reveals, zoom-through and about forty more). Three rules keep them smooth: the direction of motion carries across the cut (a zoom in is followed by a scene that starts tight and pans, not by a sudden zoom out), the same transition is never used twice in a row, and the duration is matched to

the pause in the narration at that point (short cuts on fast passages, half-second fades on calm ones).
Shorts use faster, simpler transitions than long-form.

Quality gates

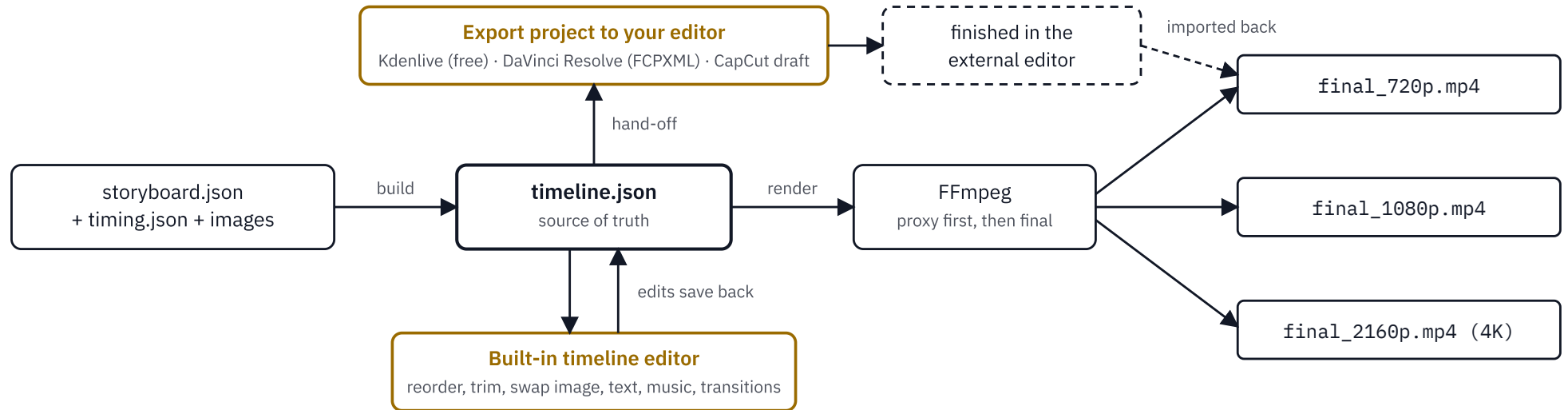
YouTube's July 2025 "inauthentic content" rule and its July 2026 clarification describe the output of a careless slideshow pipeline almost word for word. These gates turn that wording into checks the app runs itself. A failed gate blocks export until a reviewer overrides it with a logged reason.

GATE	WHAT IS CHECKED	STAGE	ON FAILURE
No advisory AI persona	Script never presents a doctor, financial adviser or lawyer giving personal advice; sensitive topics are told as stories or education with a disclaimer card	Script	block
Voice consent	Only enrolled, consented voices are used; the consent record is attached to the project	Voice	block
No competitor reuse	No competitor footage; thumbnail similarity to the reference stays under the ceiling; no logos, faces or branding copied	Export	block
Script originality	Phrase overlap and meaning similarity against the competitor transcript and the channel's recent scripts under the ceiling	Script	block
Promise kept	The title's claim and the thumbnail's subject appear in the first fifth of the video	Export	block
Visual variety	New visual at least every 8 to 12 seconds (3 to 4 for Shorts); mixed motion and transitions; share of popups and cards above the minimum; no image reused across videos	Storyboard, Edit	warn, block at threshold
Framework rotation	The same structural template is not used more than the cap per channel per 30 days	Script	warn
Disclosure	Synthetic-content flag defaults to yes; provenance metadata of images recorded, never stripped	Export	warn
Music licence	Every track in the music folder has a licence file; AI music as the main focus triggers disclosure	Edit	warn
Cadence	Uploads per channel per day stay under the configured cap; metadata templates rotate	Export	warn

Rules live in a versioned file with the source links and a "last verified" date and are re-checked quarterly, because the policy pages changed four times between July 2025 and August 2026.

Three ways a video is finished

The timeline file is the single source of truth. FFmpeg renders it automatically. For manual fixes the built-in editor changes the same file. For deeper work the project is exported to a professional editor and the finished file comes back.



Automatic renders cover most videos. The built-in editor handles the common fixes without leaving the app. The export hand-off exists for the cases that need a full professional editor.

What the built-in editor can do

AREA	ACTIONS
Scenes	Reorder by drag, split at a word, merge, trim to the nearest sentence boundary, change duration (voice pauses stretch, never the speech)
Images	Swap, regenerate with a note, upload, crop for 9:16, adjust the zoom and pan path with handles on the preview
Text	Edit popup text, position, style preset, in and out times; edit caption words and style; per-language caption tracks
Sound	Change the music track, offset, volume, ducking depth; mute a scene; re-voice one sentence
Cuts	Pick a transition and its duration from the library, or ask for a new automatic choice
Preview	Scrub with waveform, play from any point, render a low-resolution proxy of a selected range in seconds
Hand-off	Export to Kdenlive (keeps keyframes and titles), to DaVinci Resolve as FCPXML (keeps motion, captions as a separate file), to CapCut as a draft (best effort, pinned CapCut version)

Output formats and resolutions

PRESET	LONG-FORM 16:9	SHORTS 9:16	ENCODING	NOTES
720p	1280 × 720	720 × 1280	H.264, 30 fps, AAC 48 kHz	Fast preview-grade export
1080p	1920 × 1080	1080 × 1920	H.264 high profile, 30 fps, AAC 48 kHz	Default upload
4K	3840 × 2160	2160 × 3840	H.264 or H.265, 30 fps, AAC 48 kHz	Images requested at the model's largest size and scaled with a high-quality filter; slow on a laptop
Proxy	854 × 480	480 × 854	Fast H.264	For previewing cuts in the editor only

One storyboard can produce both the long-form and the Shorts version: Shorts use a 9:16 crop of each image with its own motion path, faster transitions and larger captions. Which versions to render is chosen per channel.

Languages

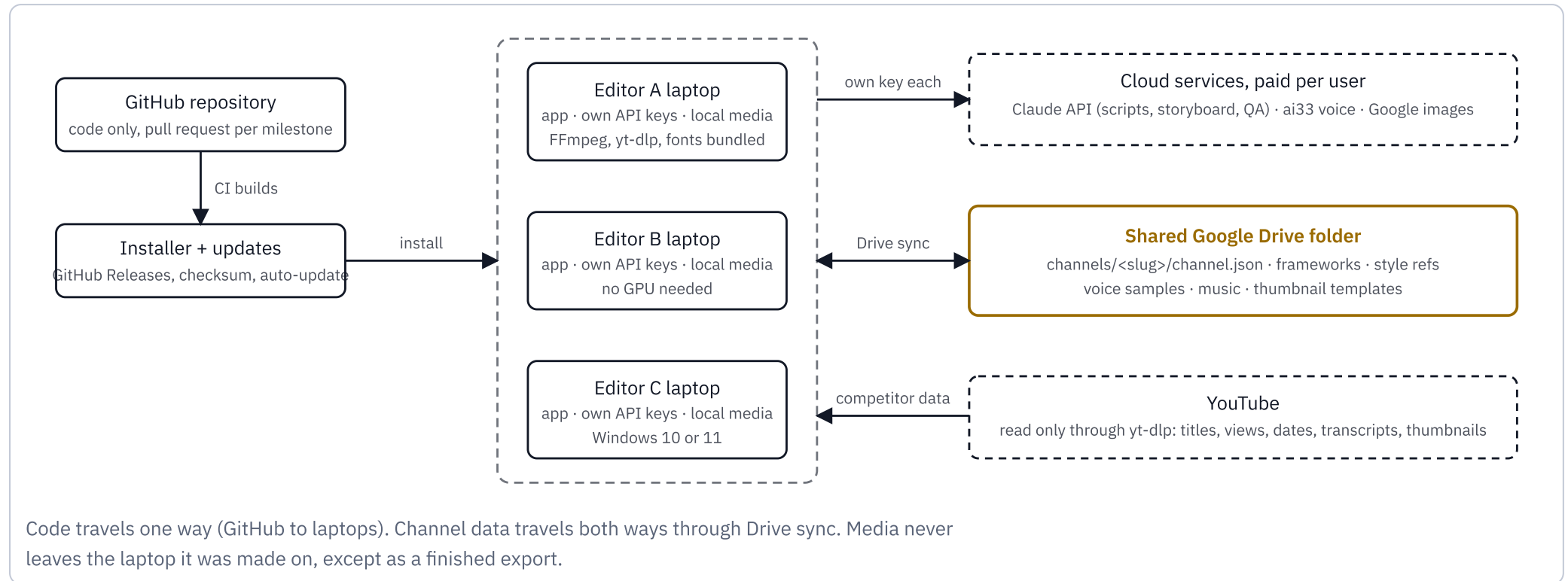
The first ten languages in order of YouTube audience size, grouped by how they earn. Script, voice, captions, popups, thumbnail text and the SEO pack are all produced in the target language.

TIER	LANGUAGES	WHY	SPECIAL HANDLING
A, high ad rates	English, German, Japanese, French, Spanish (Spain and US)	A ten-minute video pays back on tens of thousands of views	Japanese captions wrap at about 13 full-width characters; French needs non-breaking spaces before punctuation
B, volume	Hindi, Indonesian, Arabic, Spanish (Latin America), Portuguese (Brazil)	Large audiences, low ad rates; throughput matters	Hindi word timing comes from the voice tool or sentence timing (automatic alignment is unreliable); Arabic is right-to-left with proper letter joining; option to use an English master with YouTube's own dubbing instead of native production
Special	Russian	Large audience, almost no ad revenue	Only with a non-AdSense revenue plan

Fonts for every script (Latin, Devanagari, Arabic, Japanese, Korean) ship with the app so captions never fall back to boxes. A speech-normalisation pass writes out numbers, dates and currency in each language before the voice reads them.

Team setup, data and updates

No one on the team needs git or a developer tool. The installer comes from GitHub Releases and updates itself. Channel data syncs through a shared Google Drive folder. Each person uses their own API keys, stored only on their laptop.



Installing and updating

- One setup file per release, downloaded from GitHub Releases, with a checksum the app verifies before installing.
- The app checks for updates at launch and every hour; "Update now" installs and reopens the app. A bad release can be withdrawn so no one is offered it.
- FFmpeg, yt-dlp, the JavaScript runtime it needs, and all fonts are inside the install; nothing else to install by hand. The Windows web view is installed by the setup file if missing.
- The Doctor screen shows, on every laptop, what is present, what is missing, and how to fix it.

Where every file lives

Three places, with a clear rule: shared channel data in the Drive folder, per-user settings in the app's data folder, per-video media on the local drive.

Shared Google Drive folder (synced to every laptop)

channels/	
kind-ledger/	
channel.json	the Channel Setup form, saved
frameworks/	uploaded title, script, style frameworks
style/	style sheet image, reference images
thumbnail-templates/	extracted layouts from competitor thumbnails
voices/	sample recordings and consent records
music/	licensed tracks with their licence files
assets/	logos, intro and outro clips

%LOCALAPPDATA%\CashflowStudio	(per user, never shared)
settings.json	folder paths, preferences
.env	API keys for this user only
db.sqlite	job index, research cache, cost log
logs/	

Local projects folder (per user, never shared)

projects/2026-10-09_kind-ledger_why-she-paid-for-strangers/	
job.json	stage, modes, approvals, costs, provenance
01_research/	02_title/
03_script/	04_storyboard/
05_voice/	06_images/
07_edit/	08_export/
exports/kind-ledger/	finished videos, thumbnails, metadata

Keys, privacy and safety

- API keys are typed once into the Settings screen and stored only on that laptop. They are shown masked, never written to the shared folder, never included in logs or exports.
- The app's local service listens only on the laptop itself; nothing is reachable from the network.
- Channel files name the key variable to use, never the key.
- Only the team's own enrolled voices can be cloned; the consent record travels with the voice.
- YouTube access is read-only (titles, counts, dates, transcripts, thumbnails); no video is downloaded.
- Every review decision and override is logged with who, when and why, and exported in the provenance bundle for appeals.
- Spending guards: per-channel monthly budgets for voice characters and images, and a running cost per video shown on the dashboard.

Review queue and batch runs

Every stage set to review drops its result into the Review queue with a preview, the checks that ran, and three buttons: approve, edit, redo with a note. Reviewers see one queue across all channels, filtered by channel, stage and age. A batch run creates several videos at once (for example five topics for next week); each video advances on its own until it reaches a review gate, so an overnight run leaves a queue of storyboards ready to approve in the morning. Stage modes can be relaxed to auto per channel once the team trusts the output, and tightened again at any time.

Failures, retries and costs

SITUATION	WHAT THE APP DOES
YouTube asks for sign-in or rate-limits	Backs off for at least ten minutes, retries with longer pauses, then offers an optional cookie file export; research continues from the cache
Claude declines a request (safety filter)	Retries once with a reworded prompt and a fallback route; otherwise the stage waits for a person with the reason shown
Voice tool error or quota reached	Only the failed sentences are retried; cached sentences are kept; the budget guard warns before the next batch
Image fails the quality check	Regenerated up to three times with the failure reason added; then the scene is flagged for a person
Render fails	Error shown with the FFmpeg message in plain words; proxy and timeline are kept so nothing is lost
Laptop closed mid-run	Every stage writes its files before marking itself done; the job resumes from the last finished stage
Two people edit the same channel	The newer save wins and the previous version is kept in a history folder; the app shows who saved last

Cost per ten-minute video (approximate, excluding subscriptions)

ITEM	ESTIMATE	NOTES
Claude (title, script, storyboard, checks, SEO)	\$0.30 to \$0.60	Covered by the monthly API credit that comes with a Max plan, to be confirmed in the Console
Images, about 40 scenes plus thumbnail	\$1 to \$2	Google image models at \$0.03 to \$0.05 per image; half price in batch mode
Voice, about 1,500 words	depends on the ai33 plan	Per-sentence caching means edits cost only the changed sentences
Rendering, research, captions	\$0	All local

A single video as the editor sees it

1. Open the app, pick the channel. The channel already knows its competitors, frameworks, voice and image settings from the one-time setup.
2. Click New video. Research runs and shows a ranked list of competitor videos that outperformed their channel, with the outlier score, age and a thumbnail. Pick one, or type a topic.
3. Seven title variants appear with scores and reasons. Choose one or edit it.  approve.
4. The script appears in the editor with sections and the originality check result. Read it, fix anything, lock the parts you like.  approve.
5. The storyboard board shows every scene as a card: narration, image prompt, popup text, motion, transition. Change what you want, split or merge scenes.  approve.
6. Voice and images generate in the background. Images that fail the quality check are regenerated automatically; the board fills with thumbnails as they arrive. Replace or lock any image.  approve.
7. A low-resolution preview plays with captions and music. If a cut is wrong, open the timeline editor and fix it, or export the project to your own editor.
8. Final renders (720p, 1080p, 4K) land in the export folder with the thumbnail, the SEO pack and the provenance bundle.  approve, then upload.

Delivery milestones

MILESTONE	WHAT ARRIVES	DONE WHEN
M0 Foundation	App shell, Channel Setup form, Settings with keys, Doctor	A channel can be created and saved; Doctor passes
M1 Research + Title	Competitor scanning, outlier ranking, title variants, review gate	Three competitor URLs give a ranked list and an approved title
M2 Script + Storyboard	Framework ingestion, script with originality check, storyboard board	A reviewed storyboard exists for a real title
M3 Voice + Images	ai33 adapter, Google image adapter, quality check, popups	All assets generated and approved for one video
M4 Edit + Export	Timeline, FFmpeg render, presets, thumbnail, SEO pack, provenance	First complete video from competitor URL to MP4
M5 Timeline editor	Built-in editor, Kdenlive, Resolve and CapCut exports	An editor fixes a video without leaving the app
M6 Scale	Batch runs, language variants, policy dashboard, installer and auto-update	Five videos run overnight on a teammate's laptop
M7 Optional	Upload hook, analytics feedback into research	One click from export to the upload queue